

Higher-order multi-scale randomized neural network (HOMS-RNN) method for high-accuracy and efficient heat conduction simulation of composite materials

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1 Introduction

Heat conduction simulation in composite materials poses significant challenges due to highly heterogeneous microstructures with discontinuous and high-contrast material parameters. These materials exhibit complex multi-scale behaviors that are difficult to simulate using traditional numerical methods, which often require prohibitive computational resources for full fine-scale computation.

While neural network-based methods show promise for solving partial differential equations (PDEs), they face two major challenges when applied to multi-scale problems:

- **Frequency Principle:** Deep neural networks fit objective functions from low to high frequencies and exhibit slower convergence for higher frequencies, making it difficult to capture the high-frequency oscillatory behavior of multi-scale problems.
- **Prohibitive Computation:** Accurately capturing the locally oscillatory response of heterogeneous materials requires a tremendous number of sampling points, leading to significant computational challenges and potential convergence issues.

This paper introduces the Higher-Order Multi-Scale Randomized Neural Network (HOMS-RNN) method, which combines the advantages of higher-order multi-scale approaches with the efficiency of randomized neural networks to address these challenges. The method is specifically designed for transient thermal problems of composite materials with highly discontinuous and high-contrast material parameters.

2 Methodology

2.1 Multi-scale Heat Conduction Problem

The transient multi-scale heat conduction problem in composite materials is described by the following parabolic equation with periodically oscillatory and highly discontinuous coefficients:

$$\begin{cases} \rho^\varepsilon(x)c^\varepsilon(x)\frac{\partial T^\varepsilon(x,t)}{\partial t} - \frac{\partial}{\partial x_i} \left(k_{ij}^\varepsilon(x)\frac{\partial T^\varepsilon(x,t)}{\partial x_j} \right) = f(x,t), & \text{in } \Omega \times (0, T^*) \\ T^\varepsilon(x,t) = \hat{T}(x,t), & \text{on } \partial\Omega \times (0, T^*) \\ T^\varepsilon(x,0) = \hat{T}(x), & \text{in } \Omega \end{cases} \quad (1)$$

where $T^\varepsilon(x,t)$ is the temperature, $\rho^\varepsilon(x)$ is the mass density, $c^\varepsilon(x)$ is the specific heat, $k_{ij}^\varepsilon(x)$ represents the thermal conductivity coefficient, and $f(x,t)$ denotes the internal heat source. The superscript ε indicates the characteristic size of the microscopic unit cell, which causes the material parameters to be highly oscillatory and discontinuous.

2.2 Higher-Order Multi-Scale Approach

The HOMS-RNN method decouples the original multi-scale problem into simpler macroscopic and microscopic components through higher-order multi-scale analysis:

1. **Asymptotic Expansion:** The temperature solution is expanded as:

$$T^\varepsilon(x,t) = T^{(0)}(x,y,t) + \varepsilon T^{(1)}(x,y,t) + \varepsilon^2 T^{(2)}(x,y,t) + O(\varepsilon^3) \quad (2)$$

where $y = x/\varepsilon$ is the microscopic variable.

2. **Lower-Order Cell Problem:** Solving for $T^{(1)}$ leads to:

$$T^{(1)}(x,y,t) = M^{\alpha_1}(y) \frac{\partial T^{(0)}(x,t)}{\partial x_{\alpha_1}} \quad (3)$$

where $M^{\alpha_1}(y)$ is the first-order (lower-order) auxiliary unit cell function defined by:

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(y) \frac{\partial M^{\alpha_1}(y)}{\partial y_j} \right) = -\frac{\partial k_{i\alpha_1}(y)}{\partial y_i}, & \text{in } Y \\ M^{\alpha_1}(y) = 0, & \text{on } \partial Y \end{cases} \quad (4)$$

3. **Macroscopic Homogenized Problem:** The zeroth-order term $T^{(0)}$ is governed by:

$$\begin{cases} S^* \frac{\partial T^{(0)}(x,t)}{\partial t} - \frac{\partial}{\partial x_i} \left(k_{ij}^* \frac{\partial T^{(0)}(x,t)}{\partial x_j} \right) = f(x,t), & \text{in } \Omega \times (0, T^*) \\ T^{(0)}(x,t) = \hat{T}(x,t), & \text{on } \partial\Omega \times (0, T^*) \\ T^{(0)}(x,0) = \tilde{T}(x), & \text{in } \Omega \end{cases} \quad (5)$$

where the homogenized material parameters S^* and k_{ij}^* are calculated as:

$$\begin{aligned} S^* &= \frac{1}{|Y|} \int_Y \rho(y) c(y) dY \\ k_{ij}^* &= \frac{1}{|Y|} \int_Y \left(k_{ij}(y) + k_{ik}(y) \frac{\partial M^j(y)}{\partial y_k} \right) dY \end{aligned} \quad (6)$$

4. **Higher-Order Cell Problems:** The second-order term $T^{(2)}$ is defined as:

$$T^{(2)}(x, y, t) = M^{\alpha_1 \alpha_2}(y) \frac{\partial^2 T^{(0)}(x, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + R(y) \frac{\partial T^{(0)}(x, t)}{\partial t} \quad (7)$$

where $M^{\alpha_1 \alpha_2}(y)$ and $R(y)$ are second-order (higher-order) auxiliary unit cell functions defined by:

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(y) \frac{\partial M^{\alpha_1 \alpha_2}(y)}{\partial y_j} \right) = k_{\alpha_1 \alpha_2}^* - k_{\alpha_1 \alpha_2}(y) - k_{\alpha_1 j}(y) \frac{\partial M^{\alpha_2}(y)}{\partial y_j} - \frac{\partial}{\partial y_i} (k_{i \alpha_2}(y) M^{\alpha_1}(y)), & \text{in } Y \\ M^{\alpha_1 \alpha_2}(y) = 0, & \text{on } \partial Y \end{cases} \quad (8)$$

$$\begin{cases} \frac{\partial}{\partial y_i} \left(k_{ij}(y) \frac{\partial R(y)}{\partial y_j} \right) = \rho(y) c(y) - S^*, & \text{in } Y \\ R(y) = 0, & \text{on } \partial Y \end{cases} \quad (9)$$

5. **Higher-Order Multi-Scale Solution:** The final solution is assembled as:

$$T^{(2\varepsilon)}(x, t) = T^{(0)}(x, t) + \varepsilon M^{\alpha_1}(y) \frac{\partial T^{(0)}(x, t)}{\partial x_{\alpha_1}} + \varepsilon^2 \left[M^{\alpha_1 \alpha_2}(y) \frac{\partial^2 T^{(0)}(x, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + R(y) \frac{\partial T^{(0)}(x, t)}{\partial t} \right] \quad (10)$$

This approach effectively decomposes the multi-scale solution into non-oscillatory macroscopic behavior, low-frequency oscillatory information, and high-frequency oscillatory information.

2.3 Randomized Neural Networks

The HOMS-RNN method employs randomized neural networks (RNNs) to efficiently solve the separated problems:

1. **RNN Structure:** A fully connected neural network $\mathcal{N}(x; \theta)$ with depth D is defined as:

$$\begin{aligned}\varphi_0(x) &= x \\ \varphi_l(x) &= \sigma(W_l \varphi_{l-1} + b_l), \quad l = 1, \dots, D-1 \\ \mathcal{N}(x; \theta) &= \varphi_D(x) = W_D \varphi_{D-1} + b_D\end{aligned}\tag{11}$$

where $\sigma(\cdot)$ is the activation function (typically tanh).

2. **Randomization:** In an RNN, all biases and weights $[(W_1, b_1), \dots, (W_{D-1}, b_{D-1})]$ are randomly initialized from $[-R_m, R_m]$ and fixed (non-trainable). Only the weights of the last layer W_D are computed by solving a least-squares problem.
3. **Function Representation:** Any function $u_\sigma \in \mathcal{N}_\sigma^D$ (one neuron in the output layer) can be expressed as:

$$u_\sigma(x) = \sum_{i=1}^{n_{D-1}} \beta_i \varphi_i(x)\tag{12}$$

where φ_i is the i -th component of the vector φ_{D-1} and is fixed once the hidden-layer coefficients are randomly assigned.

2.4 HOMS-RNN Implementation

The HOMS-RNN method uses specialized RNNs to solve each part of the multi-scale problem:

1. **Microscopic Cell RNNs:** Two RNNs $u_\sigma^1(y)$ and $\tilde{u}_\sigma^1(y)$ are used to approximate the microscopic cell function $M^{\alpha_1}(y)$ in the matrix material Y_1 and inclusion material Y_2 :

$$u_\sigma^1(y) = \sum_{m=1}^{M_1} \beta_m^1 \varphi_m^1(y), \quad \tilde{u}_\sigma^1(y) = \sum_{m=1}^{M_2} \tilde{\beta}_m^1 \tilde{\varphi}_m^1(y)\tag{13}$$

2. **Hard-Constraint Boundary Conditions:** To improve accuracy, hard-constraint boundary conditions are applied:

$$\hat{u}_\sigma^1(y) = h(y) + \ell(y)u_\sigma^1(y) \quad (14)$$

where $h(y) = 0$ for homogeneous Dirichlet boundary conditions and $\ell(y)$ is a function that equals zero on the boundary and is positive inside the domain.

3. **Space-Time RNN:** A space-time RNN $u_\sigma^0(x, t)$ approximates the macroscopic homogenized solution $T^{(0)}(x, t)$:

$$u_\sigma^0(x, t) = \sum_{m=1}^M \beta_m^0 \varphi_m^0(x, t) \quad (15)$$

4. **Higher-Order RNNs:** Similar RNNs are used for the higher-order cell functions $M^{\alpha_1 \alpha_2}(y)$ and $R(y)$.
5. **Automatic Differentiation Assembly:** Once all components are computed, automatic differentiation is used to assemble the final HOMS-RNN solution:

$$\hat{T}^{(2\varepsilon)}(x, t) = \hat{T}^{(0)}(x, t) + \varepsilon \hat{M}^{\alpha_1}(y) \frac{\partial \hat{T}^{(0)}(x, t)}{\partial x_{\alpha_1}} + \varepsilon^2 \left[\hat{M}^{\alpha_1 \alpha_2}(y) \frac{\partial^2 \hat{T}^{(0)}(x, t)}{\partial x_{\alpha_1} \partial x_{\alpha_2}} + \hat{R}(y) \frac{\partial \hat{T}^{(0)}(x, t)}{\partial t} \right] \quad (16)$$

The algorithm can be summarized in the following steps:

1. Decouple original multi-scale PDEs into macroscopic homogenized PDEs and microscopic cell PDEs using the higher-order multi-scale model.
2. Solve lower-order microscopic cell functions using RNNs with hard-constraint boundary conditions.
3. Calculate macroscopic homogenized material parameters using the obtained microscopic cell functions.
4. Solve the macroscopic homogenized problem using a space-time RNN.
5. Solve higher-order microscopic cell functions using RNNs.
6. Assemble the final solution using automatic differentiation.

3 Theoretical Findings

The paper provides a rigorous error analysis of the HOMS-RNN method. The key theoretical results include:

Lemma 1. *Based on universal approximation theorem, for any $u \in C(I^d)$ with $I^d = [a, b]^d$ ($d = 1, 2, 3, \dots$) and $\forall \varepsilon > 0$, there exists a function $u_\sigma \in \mathcal{N}_\sigma^D$, as long as the hidden layer size is large enough, which is defined on any compact K of $C(I^d)$. Then it follows that*

$$\sup_{x \in K} |u_\sigma(x) - u(x)| < \varepsilon \quad (17)$$

Assumption 1. *For any $u \in C^3(I^d)$ and $\forall \varepsilon > 0$, there exists a function $u_\sigma \in \mathcal{N}_\sigma^D$, as long as the hidden layer size is large enough, which is defined on any compact K of $C^3(I^d)$. Then we have*

$$\max_{|\alpha| \leq 3} \sup_{x \in K} |\partial_x^\alpha u_\sigma(x) - \partial_x^\alpha u(x)| < \varepsilon \quad (18)$$

Lemma 2. *According to results in the literature, the error estimation for higher-order multi-scale solution $T^{(2\varepsilon)}(x, t)$ and exact solution $T^\varepsilon(x, t)$ of multi-scale parabolic problem is obtained as follows:*

$$\sup_{0 \leq t \leq T^*} \int_{\Omega} \left(T^\varepsilon(x, t) - T^{(2\varepsilon)}(x, t) \right)^2 dx + \int_0^{T^*} \|T^\varepsilon(x, t) - T^{(2\varepsilon)}(x, t)\|_{H^1(\Omega)}^2 dt \leq C(T^*)\varepsilon^2 \quad (19)$$

where $C(T^*)$ is a positive constant independent of ε , but dependent on total time T^* and Ω .

Theorem 1. *Based on the above lemmas and assumption, the following global error estimation is derived for the HOMS-RNN method:*

$$\sup_{0 \leq t \leq T^*} \int_{\Omega} \left(T^\varepsilon(x, t) - \hat{T}^{(2\varepsilon)}(x, t) \right)^2 dx + \int_0^{T^*} \|T^\varepsilon(x, t) - \hat{T}^{(2\varepsilon)}(x, t)\|_{H^1(\Omega)}^2 dt \leq C(T^*)\varepsilon^2 + C(T^*)\varepsilon^2 \quad (20)$$

where C is a positive constant independent of ε and ε , but dependent on total time T^* and Ω .

This theorem establishes the convergence of the HOMS-RNN method, showing that the error consists of two parts: the modeling error from the higher-order multi-scale approximation and the approximation error from the randomized neural networks.

4 Numerical Examples and Results

The paper presents several numerical experiments to demonstrate the effectiveness of the HOMS-RNN method:

4.1 Drawbacks of Direct RNN Simulation

The authors first demonstrate why direct RNN simulation is ineffective for multi-scale problems:

- As the scale parameter ε decreases (e.g., $1, 1/2, 1/3, 1/4, 1/5$), direct RNN simulations fail to accurately capture the oscillatory behavior at the micro-scale.
- Computational time increases dramatically as ε decreases (2.33s for $\varepsilon = 1$ to 1830.26s for $\varepsilon = 1/5$).
- Relative errors in direct RNN simulations are unacceptably high, especially for smaller values of ε .

4.2 Composite Materials with Different Thermal Conductivities

The HOMS-RNN method was tested on composite materials with different thermal conductivity contrasts (10:1, 50:1, and 100:1):

- **Network Configuration:** RNNs with 320 neurons in the hidden layer for microscopic cell functions and 640 neurons for macroscopic homogenized solutions; 9,000 residual points within the microscopic cell and 3,000 points along the interface.
- **Hard vs. Soft Constraints:** Hard-constraint boundary conditions showed better performance than soft constraints, with lower relative errors for all cell functions and the macroscopic solution.
- **Results for 10:1 Contrast:** The HOMS-RNN method effectively captured the microscopic oscillatory phenomena and showed stable performance throughout the temporal evolution, with errors comparable to FEM.
- **High-Contrast Materials:** For 50:1 and 100:1 contrast ratios, the lower-order multi-scale method failed to effectively capture microscopic oscillations, while the HOMS-RNN method maintained high accuracy and stability.

4.3 Robustness Analysis

The method was tested with progressively smaller characteristic parameter ε values (1/5, 1/10, 1/20, 1/30):

- The HOMS-RNN method accurately captured the oscillatory phenomena even as the number of microscopic unit cells increased dramatically.
- Computational efficiency remained stable as ε decreased, while FEM computation time increased exponentially (from 144.81s for $\varepsilon = 1/5$ to 7016.75s for $\varepsilon = 1/30$).
- The HOMS-RNN method took only 96.86s for $\varepsilon = 1/5$ and 106.46s for $\varepsilon = 1/30$, showing remarkable efficiency for large-scale simulations.

4.4 3D Composite Materials

The method was applied to 3D SiC/C composite materials:

- **Network Configuration:** RNNs with 1,280 neurons in the hidden layer; 40,000 residual points within the microscopic unit cell and 20,000 points along the interface.
- **Results:** The HOMS-RNN method accurately captured the high-frequency oscillatory response in the 3D composite, with excellent agreement with reference FEM solutions.
- **Computational Efficiency:** The 3D simulation using HOMS-RNN required only 1.5 hours, compared to 7.1 hours for the higher-order multi-scale method and 11.8 hours for reference FEM.

4.5 Porous Materials

The method was also tested on two types of porous materials:

- For both simple and complex porous geometries, the HOMS-RNN method achieved competitive accuracy with traditional numerical methods while maintaining superior computational efficiency.
- As ε decreased, computational errors of the HOMS-RNN method converged, demonstrating the robustness of the approach.

- For complex porous material, the HOMS-RNN method took 225.81s for $\varepsilon = 1/5$ and 291.85s for $\varepsilon = 1/30$, while FEM required 193.97s and 17,657.60s respectively.

5 Implications and Advantages

The HOMS-RNN method offers several significant advantages:

1. **Addressing the Frequency Principle:** By decomposing the multi-scale solution into different frequency components and using higher-order multi-scale constraints, the method effectively captures both low and high-frequency oscillatory behavior.
2. **Computational Efficiency:** The method dramatically reduces computational time compared to traditional approaches, especially for problems with small characteristic scales. The efficiency remains stable as the scale parameter decreases.
3. **Mesh-Free Advantage:** The method eliminates the need for mesh generation, which is particularly valuable for problems with complex geometries.
4. **High Accuracy:** The method achieves accuracy comparable to fine-scale FEM simulations, particularly for high-contrast materials where classical methods struggle.
5. **Space-Time Approach:** The space-time formulation for the macroscopic problem allows efficient solution of time-dependent problems without time-stepping schemes.
6. **Robustness:** The method maintains its accuracy and efficiency across different material properties, geometries, and scale parameters.

These advantages make the HOMS-RNN method particularly suitable for large-scale engineering applications involving composite materials with complex microstructures and high-contrast properties. The method's ability to efficiently handle 3D simulations and complex geometries further enhances its practical value.

Future directions for this approach include extending it to multi-physics coupling problems, leveraging parallel computing capabilities of RNNs for further efficiency improvements, and addressing nonlinear and random multi-scale problems.